# SSU Data Dash 2026: *Team name*

*Full name of every team member: First Last, First Last, ...*

**Video (YouTube, unlisted is fine):** `https://youtu.be/YOUR-VIDEO`
**Code and submission files (GitHub):** `https://github.com/YOUR-TEAM/YOUR-REPO`

## Summary

We built separate ensembles for off-market and list-aware valuation. Each combines boosted trees using property and strictly historical local-sale features with a TF–IDF text model that reads listing remarks. In six-month walk-forward validation (December 2025–May 2026), the current rerun produced the results in Table 1. Our main finding is that carefully timed local-sale history matters more than changing tree libraries: a neighbourhood feature is useful only when it is constructed from sales already known on the prediction date.

## 1 Data and features

We used physical attributes (area midpoints, lot size, bedrooms, bathrooms, parking, style, and property type), municipality and community, rounded latitude/longitude cells, calendar and seasonal variables, and indicators extracted from remarks. Local-history features summarize log price and log price per square foot for community/type/style and geographic groups. The list-aware track additionally uses list price and historical sold-to-list ratios; the off-market pipeline never receives list price. We excluded the external AVM and fields known only after sale. No outside data were used.

## 2 Models

The off-market prediction is a calibrated geometric blend of recency-aware LightGBM (32%), word TF–IDF Ridge (19%), and XGBoost (49%). The list-aware blend uses ordinary target-encoded LightGBM (6%), recency-aware LightGBM (25%), an independent off-market value anchor (13%), and a smoothed historical community sold-to-list median (56%). Trees capture nonlinear interactions among structure, location, and earlier comparable sales; Ridge contributes a different view through listing language. We blend them because their errors are not identical, even when a component is weaker by itself. The specialized list-aware anchor helps decide how much to trust an asking price by comparing it with both prior community sale ratios and a valuation that never saw list price.

We also tested a dense PyTorch MLP with standardized numeric features, property/style indicators, and learned municipality/community embeddings. It reached 11.19% off-market and 5.63% list-aware MdAPE, so we did not promote it. Its feature set lacked remarks, richer interactions, and the historical local price signals that most helped the tree ensemble; this particular architecture also fit mixed tabular structure less effectively than boosted trees.

## 3 Validation

We used walk-forward validation rather than a random split. For each month from December 2025 through May 2026, every component was rebuilt using only sales strictly before that month's cutoff, then evaluated on that month. Target encodings are also chronological *within* the training set: a training home's local statistics and global smoothing prior use only sales with an earlier `date_sold`. Homes sold on the same date are processed as a batch and cannot reveal their targets to one another. Validation homes use the complete permitted pre-cutoff history, and their sold prices never enter feature construction. Ensemble weights and calibration were selected on September–November 2025, not fitted to the six reported months.

## 4 Results

| Track | Homes | MdAPE % (95% CI) | MAPE % (95% CI) | Median signed error % |
|---|---|---|---|---|
| off-market | 10,069 | 7.27 (7.11–7.45) | 9.81 (9.62–10.00) | -1.06 |
| list-aware | 10,069 | 3.98 (3.86–4.06) | 5.19 (5.10–5.28) | -0.85 |

Table 1: Walk-forward validation on the listings. MdAPE: median absolute percentage error. MAPE: mean absolute percentage error. Brackets: bootstrap 95% interval.

The table is regenerated directly from row-level predictions by `python run.py report`. MdAPE describes the typical percentage miss, while MAPE exposes larger errors. We report both because a model can improve the median while leaving an unacceptable tail. The rerun is the authoritative score: earlier experimental numbers used leave-one-out encodings that did not meet the handout's stricter "earlier sales only" rule and are not claimed here.

## 5 Where the model fails

Errors rise in price bands and local groups with fewer comparable sales. The off-market task is consistently harder because it lacks the seller's asking price. Performance also varies by month, with later months showing sensitivity to market drift. Text helps when remarks reveal renovation or condition, but synthetic or vague descriptions add little. These patterns suggest that the next useful output is calibrated uncertainty by market segment, not just a single point estimate.

## 6 Limitations

The June–September hidden test remains the only truly unseen evaluation, so walk-forward improvement is evidence rather than a guarantee. Rare groups are still noisy despite smoothing, and recency weighting cannot anticipate a sudden market shift. The ensemble is also slower and harder to audit than a single model. December–May was inspected during development, so future tuning should use nested time splits or a newly reserved period. Next we would add segment-level uncertainty intervals and test whether simpler components can be removed without sacrificing accuracy.

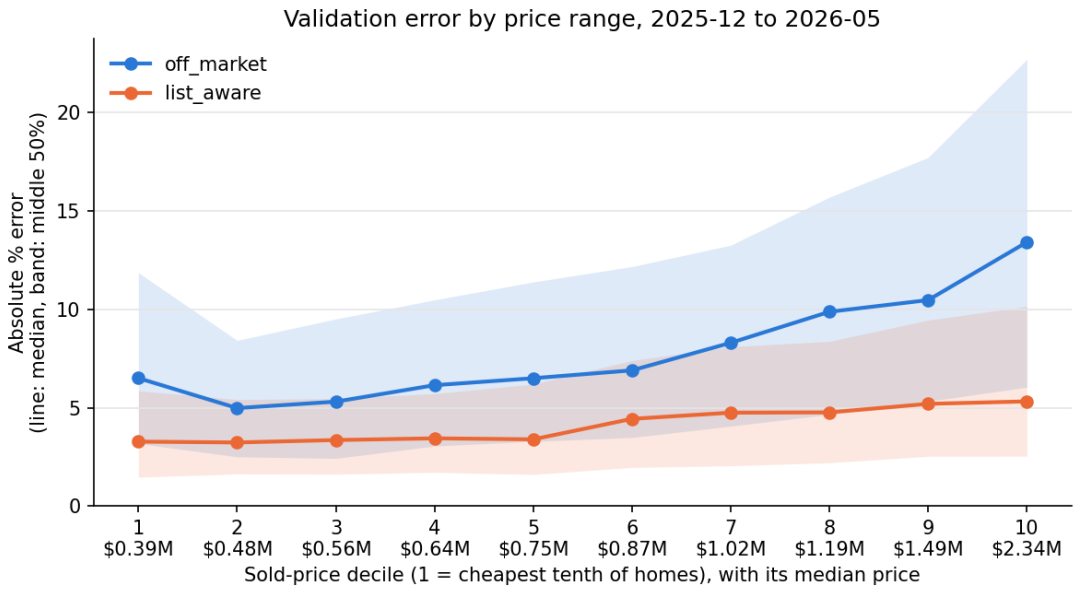



Figure 1: Absolute percentage error by sold-price decile.

## Outside data

| Source | URL | Periods covered |
|---|---|---|
| *None* | | |

## Reproducing our results

```
pip install -r gbt_ensemble_experiments/requirements-attempts.txt
python gbt_ensemble_experiments/validate.py
python run.py report
python gbt_ensemble_experiments/train_final.py
python run.py check
```